

MEDICAL CORE · RADIOTHERAPY RESEARCH

Surface-guided Respiratory Tracking (SGRT)

Build the optical part of surface-guided radiotherapy - patient-surface setup registration and respiratory gating - on a domestic 3D sensor stack.

EVIDENCE STATE Ongoing · Research	PERIOD 2026.06 - present
PORTFOLIO TIER Medical Core	PRIMARY CAPABILITY 3D Geometry & Registration

An early-stage research assignment contracted to DIGITRACK within the K-LINAC programme (led by KERI, imaging sub-project led by ETRI): far-field surface reconstruction and near-field real-time breathing tracking with commercial 3D sensors.

Problem and system boundary

01 / SURFACE TO SIGNAL

Surface to signal

Far-field sensors reconstruct the surface for planning-CT registration; near-field sensors turn chest and abdomen ROI depth into a respiratory waveform and gating signal.

PROBLEM

Patient position and breathing must be known during treatment without extra imaging or skin marks; the existing commercial system is imported

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PUBLIC BOUNDARY

First-year sensor validation only; no clinical performance or programme-target achievement is claimed, and programme targets,

Interfaces are agreed with radiation oncology, 4DCT reconstruction, image guidance, and integrated-control partners.

EVIDENCE STATE

Ongoing · Research / Measured precision of five commercial 3D sensors by distance (author-measured with the in-house tool DtDepthScan).

Owned decisions and implementation

MY ROLE

Lead the sensor validation campaign and wrote the in-house validation tool DtDepthScan (Qt, VTK, OpenCV); design and implement the breathing-tracking algorithm from ROI depth to respiratory waveform and gating signal; define the sensor interface and transport protocol; run day-to-day project work.

02 / SENSOR VALIDATION DESIGN

Sensor validation design

DtDepthScan abstracts the camera layer so five sensors run the same protocol - five distances, 500 frames

SYSTEM RELATIONSHIP

Optical surface to gating signal



Explanatory system relationship diagram; it is not photographic or experimental evidence.

IMPLEMENTATION TECHNOLOGIES

ToF camera / Structured light / Qt / VTK / OpenCV / Python / 4DCT

Verification and evidence ledger

03 / MEASURED EVIDENCE

Measured evidence

Temporal σ of ROI mean depth by distance, exposure and warm-up effects, and repeat-measurement reproducibility are tabulated.

IMAGE / PUBLICLY INSPECTABLE

Precision σ of five commercial 3D sensors vs. distance

Author-measured with the in-house validation tool DtDepthScan - silicone body phantom - 500 frames per run
 σ [mm] = temporal standard deviation of the mean ROI depth (lower is better)

Sensor	0.5 m	1 m	1.5 m	2 m	3 m	Measured fps	Fill rate
LUCID Helios2+	0.18*	0.07	0.10	0.20	0.45	24–29	$\geq 91\%$
Helios2+ Narrow	0.09*	0.05	0.10	0.10	0.18	18–25	$\geq 93\%$
Orbbec Astra 2	N/A	0.06	0.19	0.48	2.04	16–17	100%
Basler Blaze-102	0.07	0.07	0.15	0.13	0.28	~ 20	100%
Intel RealSense D455	0.15	0.41	1.90	2.99	9.93	~ 19	$\geq 91\%$

* 0.5 m values taken with reduced exposure (250 μ s) to avoid IR saturation; Astra 2 at 0.5 m is outside its minimum range (N/A).
Repeatability: ± 0.05 mm between repeated runs; σ converges within ~ 50 frames (about 2 s). ToF sensors need a 15 min warm-up (cold-start σ up to 1.7 $\times$ higher). Units: σ in mm, fps = frames per second actually delivered, fill rate = valid-depth pixels in the ROI.

Measurements by Jinmin Kim, DIGTRACK, June–July 2026 — sensor validation for surface-guided respiratory tracking (SGRT)

REGISTERED PUBLIC EVIDENCE

IMAGE / PUBLICLY INSPECTABLE

respiratory-surface-guidance-lead-01

Approved derivative; caption in portfolio data.

VERIFICATION AND EVIDENCE LEDGER

My own measurements: precision (σ), measured fps, and fill rate for five commercial 3D sensors at 0.5–3 m against a silicone body phantom, plus DtDepthScan captures.

Role, team result, and limits

04 / RESEARCH BOUNDARY

Research boundary

First-year sensor and algorithm groundwork; no clinical performance or programme-target achievement is

MY ROLE

Own sensor validation, tooling, tracking algorithm, and protocol.

Lead the sensor validation campaign and wrote the in-house validation tool DtDepthScan (Qt, VTK, OpenCV); design and implement the breathing-tracking algorithm from ROI depth to respiratory waveform and gating signal; define the sensor interface and transport protocol; run day-to-day project work.

TEAM RESULT

Consortium partners own 4DCT reconstruction, the image-guidance framework, and clinical advice; the clinical partner is the radiation

My own measurements: precision (σ), measured fps, and fill rate for five commercial 3D sensors at 0.5-3 m against a silicone body phantom, plus DtDepthScan captures.

LIMITATIONS

First-year sensor validation only; no clinical performance or programme-target achievement is claimed, and programme targets, budgets, and metrics of other institutions are not published here.

Measured precision of five commercial 3D sensors by distance (author-measured with the in-house tool DtDepthScan).

COLLABORATION BOUNDARY

Interfaces are agreed with radiation oncology, 4DCT reconstruction, image guidance, and integrated-control partners.

An early-stage research assignment contracted to DIGITRACK within the K-LINAC programme (led by KERI, imaging sub-project led by ETRI); far-field surface reconstruction and near-field real-time breathing tracking with commercial 3D

Recap and joint-development invitation

Build the optical part of surface-guided radiotherapy - patient-surface setup registration and respiratory gating - on a domestic 3D sensor stack.

PRIMARY CAPABILITY

3D Geometry & Registration / Sensor Fusion & Localization / Medical Navigation & Visualization

REGISTERED PUBLIC EVIDENCE

No external public link is currently registered.

WHAT TO BRING

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

Interfaces are agreed with radiation oncology, 4DCT reconstruction, image guidance, and integrated-control partners.

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